SIESTA Universal User Guide & First-Principles DFT Handbook

From Fundamental Quantum Mechanics to High-Throughput Calculations: Theory, FDF Architecture, Advanced Workflows, and GUI Studio Operations

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Computational Materials Laboratory & Surface Engineering Group
Interactive Web Studio: https://chemsiva.github.io/siesta-input-builder/

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Target Codebases:
`SIESTA 5.x (5.4.2)` & `SIESTA Input Builder Web Studio`

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Online Interactive GUI: https://chemsiva.github.io/siesta-input-builder/
Repository: https://github.com/chemsiva/siesta-input-builder

This handbook is designed as a universal primer: a complete beginner with zero prior knowledge of density functional theory or SIESTA can master the underlying physics, parameters, and input architecture from the ground up, while experienced computational researchers can leverage the detailed operational reference, general constraints (Section 7.7), time accounting (Section 6.29), and converter pipelines.

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Foundations of SIESTA & First-Principles Physics

The Quantum Many-Body Challenge and Hohenberg-Kohn Theorems

In condensed matter physics and quantum chemistry, the fundamental goal is to solve the time-independent Schrödinger equation for a collection of N interacting electrons and M atomic nuclei:

$$\hat{H}\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N; \mathbf{R}_1, \ldots, \mathbf{R}_M) = E\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N; \mathbf{R}_1, \ldots, \mathbf{R}_M)$$

Under the **Born-Oppenheimer approximation**, because atomic nuclei are thousands of times more massive than electrons ($M_{\text{nuc}} \gg m_e$), nuclear kinetic energy is decoupled, treating nuclei as static external point charges that produce a Coulomb potential $V_{\text{ext}}(\mathbf{r})$ felt by the electrons.

Even with this simplification, the exact electronic wave function $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ depends on $3N$ spatial coordinates. For a modest semiconductor slab containing 100 atoms (~1000 valence electrons), Ψ lives in a 3000-dimensional Hilbert space, rendering direct wave function approaches computationally impossible for solid-state materials.

Density Functional Theory (DFT) circumvents this exponential bottleneck through the two foundational theorems of **Hohenberg and Kohn (1964)**:

1. **Theorem 1 (One-to-One Mapping):** The ground-state electron density $\rho(\mathbf{r})$ uniquely determines the external potential $V_{\text{ext}}(\mathbf{r})$ up to an additive constant, and therefore uniquely determines the full Hamiltonian and all ground-state observables of the many-body system.

2. **Theorem 2 (Variational Principle):** There exists a universal energy functional $E[\rho]$ such that the exact ground-state density minimizes $E[\rho]$, yielding the true ground-state energy E_0:

$$E_0 \leq E[\tilde{\rho}] \quad \text{for any trial density } \tilde{\rho}(\mathbf{r}) \geq 0, \quad \int \tilde{\rho}(\mathbf{r}) d^3\mathbf{r} = N$$

The Kohn-Sham Formulation

To make this variational principle computationally tractable, **Kohn and Sham (1965)** replaced the intractable interacting electron system with an auxiliary, non-interacting electron system that yields the exact same ground-state density $\rho(\mathbf{r})$. The total energy functional is partitioned into four physically transparent terms:

$$E[\rho] = T_s[\rho] + \int \rho(\mathbf{r}) V_{\text{ext}}(\mathbf{r}) d^3\mathbf{r} + E_H[\rho] + E_{\text{xc}}[\rho]$$

where:

- $T_s[\rho] = -\frac{\hbar^2}{2m}\sum_{i=1}^{N}\int \psi_i^*(\mathbf{r})\nabla^2\psi_i(\mathbf{r})d^3\mathbf{r}$ is the kinetic energy of non-interacting electrons described by single-particle Kohn-Sham orbitals $\psi_i(\mathbf{r})$.
- $E_H[\rho] = \frac{e^2}{2}\iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d^3\mathbf{r}d^3\mathbf{r}'$ is the classical electrostatic Hartree electron-electron repulsion.
- $E_{\text{xc}}[\rho]$ is the **Exchange-Correlation (XC) functional**, which encapsulates all quantum mechanical many-body effects: Pauli exchange repulsion and dynamic electron correlation.

Varying $E[\rho]$ with respect to $\psi_i^*(\mathbf{r})$ yields the canonical **Kohn-Sham Equations**:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{eff}}[\rho](\mathbf{r})\right)\psi_i(\mathbf{r}) = \varepsilon_i\psi_i(\mathbf{r}), \quad \text{where } \rho(\mathbf{r}) = \sum_{i=1}^{N}|\psi_i(\mathbf{r})|^2$$

with the effective single-particle potential defined as:

$$V_{\text{eff}}(\mathbf{r}) = V_{\text{ext}}(\mathbf{r}) + e^2\int\frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d^3\mathbf{r}' + \frac{\delta E_{\text{xc}}[\rho]}{\delta\rho(\mathbf{r})}$$

Because V_{eff} depends on $\rho(\mathbf{r})$, which in turn depends on the unknown wavefunctions $\psi_i(\mathbf{r})$, the Kohn-Sham equations must be solved iteratively until self-consistency is achieved (**Self-Consistent Field, SCF**).

Plane-Wave DFT vs. Linear Combination of Atomic Orbitals (LCAO)

To solve the differential Kohn-Sham equations on a computer, the continuous orbitals $\psi_i(\mathbf{r})$ must be expanded in a discrete mathematical basis set:

$$\psi_i(\mathbf{r}) = \sum_{\mu} c_{\mu i}\phi_\mu(\mathbf{r})$$

The choice of basis functions $\phi_\mu(\mathbf{r})$ fundamentally distinguishes computational DFT codes:

Comparison between plane-wave DFT methods and the localized atomic orbital approach in SIESTA.

Feature	Plane-Wave Codes (QE, VASP)	LCAO Codes (SIESTA)
Basis Function ϕ_μ	Orthogonal Fourier waves: $e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}$	Strictly localized numerical atomic orbitals: $R_{nl}(r)Y_{lm}(\hat{r})$
Systematic Convergence	Trivial: increase energy cutoff E_{cut}	Requires selecting radial cutoffs and polarization shells (SZ, DZP, TZP)
Vacuum Efficiency	Very poor: empty vacuum space requires plane waves everywhere	**Extremely efficient**: empty vacuum has zero orbitals $\Longrightarrow$ zero compute cost
Scaling with Size N	Cubic $O(N^3)$ or $O(N^2 \log N)$ FFT cost	**Linear $O(N)$ or sparse $O(N)$ Hamiltonian**
Sparsity	Dense matrices in real space	**Strictly sparse Hamiltonian & Overlap matrices**
Ideal Applications	Dense 3D metals, isotropic bulk solids	**Surface slabs, 2D layers, large clusters, nanoribbons, biomolecules**

Why SIESTA? The $O(N)$ Localized Orbital Paradigm

SIESTA (*Spanish Initiative for Electronic Simulations with Thousands of Atoms*) was conceived by Emilio Artacho, Alberto García, Julian Gale, Javier Junquera, Richard Martin, Pablo Ordejón, Daniel Sánchez-Portal, and José M. Soler.

The defining breakthrough of SIESTA is that its basis orbitals $\phi_\mu(\mathbf{r})$ are **strictly zero** beyond a finite cutoff radius $r_{c,\mu}$:

$$\phi_\mu(\mathbf{r}) \equiv 0 \quad \text{for } |\mathbf{r} - \mathbf{R}_I| > r_{c,\mu}$$

Because orbitals strictly vanish beyond r_c, the overlap matrix $S_{\mu\nu} = \langle \phi_\mu | \phi_\nu \rangle$ and Hamiltonian matrix $H_{\mu\nu} = \langle \phi_\mu | \hat{H} | \phi_\nu \rangle$ are **sparse banded matrices**: matrix elements are strictly zero unless two atoms are within distance $r_{c,\mu} + r_{c,\nu}$. As a direct consequence:

- Matrix construction scales strictly as $\mathcal{O}(N)$.
- Storing matrices requires memory proportional to $\mathcal{O}(N)$ rather than $\mathcal{O}(N^2)$.
- Systems comprising thousands of atoms (e.g. $FAPbI_3$ perovskite slabs with organic ligands) can be calculated on standard computing clusters where plane-wave codes would run out of memory.

The FDF (Flexible Data Format) Language

Syntax, Tokens, and Data Blocks

All control parameters, structural descriptions, and calculation directives in SIESTA are provided through a unified plain-text file known as the **FDF (Flexible Data Format)** file (conventionally named with the `.fdf` extension).

Key rules of the FDF format:

1. **Case Insensitivity:** FDF keywords are case-insensitive. `MeshCutoff`, `meshcutoff`, and `Mesh.Cutoff` resolve to the exact same parameter.
2. **Punctuation Tolerance:** Characters like periods (`.`), hyphens (`-`), and underscores (`_`) in variable names are largely treated as interchangeable separators in modern SIESTA versions.
3. **Single Values:** Specified as `Keyword Value [Unit]`:

```
MeshCutoff         350.0 Ry
ElectronicTemperature  25.0 meV
```

4. **Logical Values:** Booleans can be specified using `true` / `false`, `yes` / `no`, or single letters `T` / `F`.
5. **Data Blocks:** Structured arrays, coordinate tables, and matrix specifications are enclosed in `%block BlockName` and `%endblock BlockName`:

```
%block ChemicalSpeciesLabel
  1   14  Si      Si.psf
%endblock ChemicalSpeciesLabel
```

6. **Comments:** Any text preceded by `#` or `!` is treated as a comment and ignored by the parser.

Physical Dimensions and Unit Scaling

SIESTA provides automatic physical unit conversion. A user can specify units in their preferred scientific convention, and SIESTA internally scales them to atomic units (Hartree, Bohr):

Supported physical units and dimensional scaling in SIESTA FDF files.

Physical Quantity	Supported FDF Units	Common Conversions
Length	`Ang`, `Bohr`, `nm`, `pm`	1 Å = 10^{-10} m ≈ 1.889726 Bohr
Energy	`eV`, `Ry`, `Hartree`, `kcal/mol`, `kJ/mol`, `meV`	1 Ry = 13.60569 eV = 0.5 Hartree
Pressure / Stress	`GPa`, `kBar`, `bar`, `Pa`, `Ry/Bohr**3`, `eV/Ang**3`	1 GPa = 10 kBar = 0.0062415 eV/Å^3
Force	`eV/Ang`, `Ry/Bohr`, `Hartree/Bohr`	1 eV/Å ≈ 0.036749 Ry/Bohr
Temperature	`K`, `eV`, `meV`, `Ry`	1 eV ≈ 11604.5 K, 300 K ≈ 25.85 meV
Time	`fs`, `ps`, `s`, `mins`, `hours`, `days`	1 fs = 10^{-15} s

Crystal Structure, Lattice & Geometry Setup

Lattice Representation: Vectors vs. Parameters

Every periodic calculation requires defining the unit cell Bravais lattice. In SIESTA, this is governed by the global scaling parameter `LatticeConstant` and one of two mutually exclusive blocks:

Option A: Explicit Lattice Vectors (Recommended)

```
LatticeConstant 1.0 Ang
%block LatticeVectors
  12.640000   0.000000   0.000000
   0.000000  12.640000   0.000000
   0.000000   0.000000  36.000000
%endblock LatticeVectors
```

Here, each row represents the Cartesian components of basis vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. Because `LatticeConstant` is set to `1.0 Ang`, the numbers inside the block represent absolute Cartesian lengths in Ångströms.

Option B: Crystallographic Lattice Parameters

```
LatticeConstant 5.4307 Ang
%block LatticeParameters
  1.0  1.0  1.0  90.0  90.0  90.0
%endblock LatticeParameters
```

The row defines $a/a_0, b/a_0, c/a_0, \alpha, \beta, \gamma$, where α is the angle between $\mathbf{a}_2$ and $\mathbf{a}_3$, β between $\mathbf{a}_1$ and $\mathbf{a}_3$, and γ between $\mathbf{a}_1$ and $\mathbf{a}_2$.

Atomic Coordinates Formats

The spatial positions of all atoms in the simulation cell are declared in the `%block AtomicCoordinatesAndAtomicSpecies` block. The interpretation of these coordinates is dictated by `AtomicCoordinatesFormat`:

- `Fractional`: Dimensionless crystal coordinates (u, v, w) such that position $\mathbf{R} = u\mathbf{a}_1 + v\mathbf{a}_2 + w\mathbf{a}_3$, with $0 \le u, v, w < 1$.

- `NotScaledCartesianAng` : Unscaled Cartesian coordinates (x, y, z) in Ångströms.
- `NotScaledCartesianBohr` : Unscaled Cartesian coordinates in atomic units (Bohr).
- `ScaledCartesian` : Cartesian coordinates multiplied by the global `LatticeConstant` .

```
AtomicCoordinatesFormat Fractional
%block AtomicCoordinatesAndAtomicSpecies
  0.00000000  0.00000000  0.15000000  1   # Pb atom at bottom
  0.50000000  0.50000000  0.15000000  2   # I atom
%endblock AtomicCoordinatesAndAtomicSpecies
```

The fourth column is the integer species index corresponding to the sequential index in `%block ChemicalSpeciesLabel` .

Surface Slabs, Vacuum Padding, and Dipole Correction

When modeling surfaces, interfaces, or 2D materials (graphene, perovskite nanosheets, transition metal dichalcogenides), three-dimensional periodic boundary conditions (PBC) are still enforced by the FFT grid. To prevent spurious electronic interactions between periodic vertical images of the slab along the surface normal (z), one must:

1. **Add Sufficient Vacuum:** Typically 15 Å $\le d_{\text{vac}} \le$ 25 Å between the highest atom of one slab and the lowest atom of the periodic replica.
2. **Center the Slab:** Shift atomic coordinates so the vacuum is evenly split above and below the material (`AtomicCoordinatesOrigin COP-Z`).
3. **Apply Dipole Correction:** If the slab is asymmetric (e.g., polar surface termination or molecule adsorbed on one side only), an artificial net dipole moment **p** arises across the cell. This induces a fictitious linear electric field across the vacuum. Setting:

   ```
   Slab.DipoleCorrection true
   ```

   instructs SIESTA to introduce an internal compensating dipole sheet in the vacuum gap, flattening the potential and yielding exact, uncorrupted work functions.

Pseudopotentials & PAO Basis Sets

The Pseudopotential Concept (Norm-Conservation & Kleinman-Bylander)

Core electrons (e.g. $1s^2 2s^2 2p^6$ in Silicon) are tightly bound to the nucleus and chemically inert. However, their wavefunctions oscillate violently near the core to maintain orthogonality with deeper shells. In pseudopotential theory:

1. Core electrons and the nuclear Coulomb potential are replaced by an effective smooth **pseudopotential** $V_{\text{pseudo}}(r)$.
2. Beyond a core radius r_c, the pseudo-wavefunction $\tilde{\psi}(r)$ matches the all-electron wavefunction $\psi(r)$ exactly.
3. Under **norm-conservation**, the integral of $|\tilde{\psi}(r)|^2$ inside r_c equals that of the all-electron wavefunction, ensuring correct electrostatic scattering properties across varying chemical environments.

SIESTA requires norm-conserving pseudopotentials in the separable **Kleinman-Bylander (KB)** form:

$$\hat{V}_{\mathrm{NL}} = \sum_{l,m} \frac{|\chi_{lm}\rangle\langle\chi_{lm}|}{\langle\chi_{lm}|\delta V_l|\chi_{lm}\rangle}$$

Pseudopotential File Formats: PSF and Modern PSML

- **PSF (Traditional ASCII Format):** Stores radial grid data, logarithmic derivatives, and local/non-local projectors. Named `[Element].psf` (e.g., `Pb.psf`, `I.psf`).
- **PSML (Pseudopotential Markup Language):** The modern XML-based standard adopted by SIESTA 5.x. Enables full provenance tracking, relativistic spin-orbit projectors, and semicore configuration verification.

Numerical Atomic Orbitals (NAOs) and Basis Cardinality

Unlike Gaussian orbitals used in quantum chemistry, SIESTA computes radial basis functions numerically by solving the Schrödinger equation for the isolated pseudo-atom enclosed in a spherical confinement well:

$$\phi_{nlm}(\mathbf{r}) = R_{nl}(r)Y_{lm}(\hat{\mathbf{r}})$$

The basis set hierarchy is defined by `PAO.BasisSize`:

- `SZ` **(Single-Zeta):** Minimal basis. Exactly one radial function per valence orbital (e.g. for Carbon $2s, 2p$: 1 s function and 3 p functions = 4 basis functions). Fast, qualitative.
- `DZ` **(Double-Zeta):** Two radial functions per valence orbital ($2s, 2s', 2p, 2p'$). Essential for chemical bonding and valence deformation.
- `DZP` **(Double-Zeta Polarized): The standard for production research.** Takes `DZ` and adds a polarization shell with angular momentum $l+1$ (e.g. adds $3d$ orbitals to Carbon $2p$). Allows electron clouds to distort under asymmetric surface fields.
- `TZP` / `TZ2P`**:** Triple-zeta with multiple polarization shells. High-precision limit, approaching the complete basis set (CBS) plane-wave accuracy.

The Spatial Confinement Parameter: `PAO.EnergyShift`

How does SIESTA determine where to truncate the orbital so it strictly equals zero at r_c? Instead of forcing the user to guess radii for every shell of every element, SIESTA provides the universal thermodynamic parameter `PAO.EnergyShift`:

$$\delta E = \frac{\hbar^2 \pi^2}{2 m_e r_{c,nl}^2}$$

- **Physical Meaning:** It represents the energy increase experienced by an electron when it is confined inside a spherical box of radius r_c.
- **Standard Value:** Default is `100 meV` to `250 meV`.
- **Behavior:** A *smaller* energy shift (e.g. `50 meV`) gives a larger r_c, expanding the basis set to capture diffuse vacuum states at the cost of denser matrices. A *larger* energy shift (e.g. `300 meV`) makes orbitals more compact and faster, but may under-represent surface dipole relaxation.

Real-Space Grid & Exchange-Correlation (XC)

The Real-Space Mesh and `MeshCutoff`

While Hamiltonian and overlap matrices are computed on atomic basis orbitals, the electron density $\rho(\mathbf{r})$ and Hartree/XC potentials are represented on a 3D regular Cartesian real-space grid. The grid fineness is specified by `MeshCutoff`:

$$E_{\text{cut}} = \frac{\hbar^2 k_{\max}^2}{2m_e} \Longrightarrow \Delta x \approx \frac{\pi}{k_{\max}} = \frac{\pi}{\sqrt{2m_e E_{\text{cut}}}}$$

- For $E_{\text{cut}} = 200\ \text{Ry} \Longrightarrow \Delta x \approx 0.12\ \text{Å}$.
- For $E_{\text{cut}} = 350\ \text{Ry} \Longrightarrow \Delta x \approx 0.09\ \text{Å}$.
- **Rule of Thumb:** Always perform a mesh cutoff convergence test. Standard semiconductors require 250–350 Ry. Highly localized elements (O, F, Fe) require 350–450 Ry.

The Eggbox Effect

Because the real-space grid is discrete and does not automatically displace with the atoms as they move, shifting an atom across grid points induces an artificial periodic ripple in the total energy and forces, termed the **eggbox effect**:

- **Remedy 1:** Increase `MeshCutoff` (> 300 Ry).
- **Remedy 2:** Use modern grid-cell sampling (`GridCellSampling true`).
- **Remedy 3:** Use soft confinement potentials (`PAO.SoftDefault true`) to smooth the orbital second derivatives near r_c .

Exchange-Correlation (XC) Functionals

SIESTA supports both local and gradient-corrected approximations:

```
XC.functional   GGA
XC.authors      PBE
```

- **LDA (Local Density Approximation):** `XC.functional LDA` , `XC.authors CA` (Ceperley-Alder parameterized by Perdew-Zunger). Reliable for bulk moduli, underestimates lattice parameters by 1–2%.
- **GGA (Generalized Gradient Approximation):** `XC.functional GGA` , `XC.authors PBE` (Perdew-Burke-Ernzerhof). The workhorse functional for solid-state chemistry and surfaces.
- **PBEsol:** `XC.authors PBEsol` . Refined GGA specifically tuned for equilibrium lattice constants and bulk phonon frequencies.
- **van der Waals Dispersion (Grimme DFT-D3):** Crucial for molecular adsorption, perovskite organic-inorganic interfaces, and layered materials:

  ```
  DFTD3           true
  ```

Spin Polarization, Non-Collinear Magnetism & Spin-Orbit Coupling

SIESTA provides three levels of magnetic treatment:

1. **Collinear Spin Polarized:** α and β spins along a single quantization axis:

```
Spin            polarized
```

2. **Non-Collinear Spin:** Vector spin density $\mathbf{S}(\mathbf{r}) = (S_x, S_y, S_z)$ pointing arbitrarily in 3D space:

```
Spin            non-collinear
```

3. **Fully Relativistic Spin-Orbit Coupling (SOC):** Essential for heavy elements (Lead Pb, Iodine I, Bismuth Bi, Gold Au):

```
Spin            spin-orbit
```

Self-Consistent Field (SCF) Control

The Density Matrix Formulation

Instead of dealing with individual Kohn-Sham wavefunctions, SIESTA expresses all physical observables via the single-particle **density matrix** $P_{\mu\nu}$:

$$P_{\mu\nu} = \sum_i f_i c_{\mu i} c^*_{\nu i}$$

where f_i is the Fermi-Dirac occupation factor. The real-space electron density is reconstructed as:

$$\rho(\mathbf{r}) = \sum_{\mu,\nu} P_{\mu\nu} \phi^*_\mu(\mathbf{r}) \phi_\nu(\mathbf{r})$$

SCF Convergence Criteria and Mixing Algorithms

During each SCF iteration m, the computed density matrix $P^{(m)}$ differs from the input density matrix $P^{(m)}_{\text{in}}$. A simple substitution $P^{(m+1)}_{\text{in}} = P^{(m)}$ leads to violent charge sloshing and divergence. Therefore, mixing algorithms are applied:

```
MaxSCFIterations      150
DM.Tolerance          1.0d-4
DM.MixingWeight       0.25
DM.NumberPulay        8
```

- **`DM.Tolerance`:** The calculation is declared converged when:
$$\max_{\mu,\nu} |P^{(m)}_{\mu\nu} - P^{(m-1)}_{\mu\nu}| < \texttt{DM.Tolerance} \quad \text{(typically } 10^{-4} \text{ to } 10^{-5}\text{)}$$
- **`DM.MixingWeight` (α):** Fraction of the newly calculated density matrix mixed with the old matrix. For difficult systems (surfaces, transition metals, low bandgap perovskites), lower this value to `0.05` or `0.10` to prevent divergence.
- **`DM.NumberPulay`:** Depth of the Pulay Direct Inversion in the Iterative Subspace (DIIS) history. Using 6 to 8 history steps accelerates convergence exponentially after the initial linear regime.

Electronic Temperature (Fermi Smearing)

For metallic systems or surfaces with near-zero bandgaps, sharp discontinuities in Fermi occupations at $T = 0$ destabilize SCF convergence. Introducing a finite electronic temperature smears occupations via the Fermi-Dirac function:

$$f(\varepsilon) = \frac{1}{e^{(\varepsilon - E_F)/k_B T_{\text{elec}}} + 1}$$

```
ElectronicTemperature  25.0 meV
```

This represents $T \approx 290$ K, smoothing occupation transitions and ensuring rapid, monotonic convergence.

Brillouin Zone Sampling & K-Points

Reciprocal Space Integration & Monkhorst-Pack Grids

In crystalline solids, translational symmetry dictates that wavefunctions obey Bloch's theorem: $\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$. Physical properties require integration over the first Brillouin zone:

$$\bar{g} = \frac{\Omega}{(2\pi)^3} \int_{\text{BZ}} g(\mathbf{k}) d^3\mathbf{k} \approx \sum_{\mathbf{k}} w_{\mathbf{k}} g(\mathbf{k})$$

SIESTA generates discrete k-point grids using the **Monkhorst-Pack (MP)** scheme:

```
%block kgrid_Monkhorst_Pack
  4   0   0   0.5
  0   4   0   0.5
  0   0   1   0.0
%endblock kgrid_Monkhorst_Pack
```

The first three numbers in each row define the divisions along reciprocal vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$. The fourth number is the origin shift (typically `0.5` to avoid high-symmetry points or `0.0` for Γ-centered meshes).

Dimensionality Rules for K-Points

- **3D Bulk Crystals (e.g. Silicon, Perovskite bulk):** Full 3D grid ($6 \times 6 \times 6$ or $8 \times 8 \times 8$).
- **2D Surfaces & Slabs: Strictly 1 point along** z ($k_z = 1$, e.g. $4 \times 4 \times 1$). Because the slab has vacuum along z, there is zero physical dispersion along the reciprocal z-axis. Setting $k_z > 1$ wastes massive compute power with zero physical benefit.
- **1D Nanowires / Nanotubes:** 1 point along x and y ($1 \times 1 \times 8$).
- **0D Molecules / Isolated Clusters:** Single Γ-point calculation ($1 \times 1 \times 1$).

Structural Relaxation, General Constraints & Molecular Dynamics

Geometry Optimization Algorithms

Structural relaxation minimizes the total energy with respect to nuclear positions and cell vectors until atomic forces and stress tensors fall below strict tolerances:

```
MD.TypeOfRun        CG
MD.NumCGsteps       200
MD.MaxForceTol      0.04 eV/Ang
MD.MaxStressTol     1.0 GPa
MD.VariableCell     false
```

- **`MD.TypeOfRun`:** Options include `CG` (Conjugate Gradient, standard), `Broyden` (quasi-Newton), and `FIRE` (Fast Inertial Relaxation Engine, superior for complex interfaces and loose initial configurations).
- **`MD.MaxForceTol`:** The maximum force component allowed on any atom for convergence (typically 0.02–0.04 eV/Å).
- **`MD.VariableCell`:** Set to `true` to relax cell vectors simultaneously. **Critical Warning:** Never enable `MD.VariableCell true` on a surface slab without fixing the *c*-axis, or the simulation will compress the vacuum gap!

SIESTA Section 7.7: Use of General Constraints

SIESTA provides a versatile constraints engine (`%block Geometry.Constraints`) to freeze or couple degrees of freedom during relaxation:

```
%block Geometry.Constraints
  atom [1 -- 12]              # Freeze atoms 1 through 12 completely
  atom 15  0.  0.  1.         # Fix atom 15 along Z only (allow XY relaxation)
  cell-vector C               # Freeze out-of-plane vacuum axis
  stress XX YY                # Constrain in-plane stress components
  Z 82                        # Freeze all Lead (Pb) atoms by atomic number
%endblock Geometry.Constraints
```

Key Constraint Syntax Rules

- **Range specifications:** Supports `atom [1 – 4]`, `atom from 1 to 4 step 1`, and space-separated lists `atom 1 2 3 4`.
- **Directional Constraints:** Append three real numbers representing a Cartesian projection vector. **Crucial Rule:** Reals *must* include a trailing decimal point (e.g., `0. 0. 1.`). SIESTA subtracts the force component projected onto this vector:

$$\mathbf{F}_{\text{active}} = \mathbf{F} - (\mathbf{F} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$$

- **Cell Constraints:** `cell-vector C` locks the vacuum height, allowing in-plane cell vectors A and B to relax without collapsing the vacuum.

Post-Processing, Spectroscopy & Performance Accounting

Band Structure Calculations (`%block BandLines`)

Electronic dispersion curves $E_n(\mathbf{k})$ along high-symmetry Brillouin zone paths:

```
BandLinesScale ReciprocalLatticeVectors
%block BandLines
  1   0.000  0.000  0.000  \Gamma
 25   0.500  0.000  0.500  X
 15   0.500  0.250  0.750  W
 20   0.500  0.500  0.500  L
 25   0.000  0.000  0.000  \Gamma
%endblock BandLines
```

SIESTA writes `SystemLabel.bands`, which can be plotted directly using the web studio or converted via `gnubands SystemLabel.bands > bands.dat`.

Projected Density of States (PDOS)

Orbital-resolved density of states $g_\mu(E) = \sum_{n,\mathbf{k}} w_\mathbf{k} |\langle \phi_\mu | \psi_{n\mathbf{k}} \rangle|^2 \, \delta(E - E_{n\mathbf{k}})$:

```
%block ProjectedDensityOfStates
  EF  -10.0   10.0   0.100   1000   eV
%endblock ProjectedDensityOfStates
```

Outputs `SystemLabel.PDOS.xml`, decomposing contributions into atomic species, angular momentum shells (s, p, d, f), and magnetic orbital orientations $(p_x, p_y, p_z, d_{z^2}, d_{x^2-y^2})$.

Work Function Determination ($\Phi = V_{\text{vac}} - E_F$)

The work function Φ is the minimum energy required to remove an electron from the Fermi level of a solid to the electrostatic vacuum plateau outside the surface:

$$\Phi = V_{\text{vac}} - E_F$$

1. Request the electrostatic potential grid: `SaveElectrostaticPotential true` $\Longrightarrow$ generates `SystemLabel.VH`.
2. Compute planar average $\bar{V}(z) = \frac{1}{A}\iint V(\mathbf{r}) dx dy$ using `Util/Macroave` or `grid2cube`.
3. Identify the asymptotic flat plateau V_{vac} in the middle of the vacuum region.
4. Subtract the self-consistent Fermi level: $\Phi = V_{\text{vac}} - E_F$.

Section 6.29: Memory, CPU-time & Wall-Time Accounting

Large-scale HPC calculations require monitoring memory and execution bottlenecks:

```
AllocReportLevel         2           # Detailed memory report in SystemLabel.alloc
AllocReportThreshold     1000000     # Report arrays larger than 1 MB
TimerReportThreshold     0.001       # Print routines taking > 0.1% CPU in TIMES
UseParallelTimer         true        # Write MPI timing breakdown to SystemLabel.times
MaxWalltime              24 hours    # Queue time ceiling before clean abort
MaxWalltime.Slack        10 s        # Emergency shutdown buffer
```

The SIESTA Input Builder Web Studio Manual

Application Overview & Zero-Backend Architecture

The **SIESTA Input Builder** is a pure client-side web application running entirely inside the modern browser via React 19, Vite, and Zustand.

Core Advantages:

- **Zero Server Compute & Total Privacy:** Input building, CIF parsing, and data conversion execute strictly in local browser memory. No proprietary crystal coordinates, structures, or logs are uploaded to any external server.
- **Real-Time Bi-Directional Preview:** Any keystroke in the form immediately updates the live, syntax-colored `.fdf` code preview panel.
- **Cross-Code Ecosystem:** Includes specialized converters to translate between VASP POSCAR (with ASE GUI Selective Dynamics), CIF crystallographic files, and SIESTA.

Walkthrough of the Navigation Modules

1. **1 System:** Sets `SystemLabel`, `SystemName`, and optional atom/species counts.
2. **2 Structure:** Full interactive coordinate table, crystallographic unit cell parameters, vacuum cleavage tools, supercell expansion, and 3D WebGL Crystal Viewer.
3. **3 Species & Basis:** Defines chemical elements, links pseudopotential files (`.psf` / `.psml`), sets basis cardinality (`SZ`, `DZP`, `TZP`), and configures `PAO.EnergyShift`.
4. **4 DFT / XC:** Exchange-correlation selection (LDA, GGA-PBE, PBEsol), `MeshCutoff`, Grimme DFT-D3, spin polarization, and DFT+U.
5. **5 SCF Control:** Iteration limits, density matrix tolerances, Pulay mixing history, and electronic smearing temperature.
6. **6 K-Points:** Monkhorst-Pack reciprocal grid configuration with automated dimensional checks ($k_z = 1$ for surfaces).
7. **7 Bands & DOS / PDOS:** High-symmetry Brillouin zone path generation and projected DOS energy window controls.
8. **8 Output Flags & Accounting:** Output toggles (`.ANI`, `.MDE`, `.RHO`, `.VH`) and Section 6.29 `TIMES` / `alloc` execution profiler.
9. **9 MD / Relaxation:** Structural optimizers (CG, FIRE), force tolerances, variable cell relaxations, and molecular dynamics ensembles.

10. **1 0 CI-NEB Suite:** Climbing Image Nudged Elastic Band transition state search, IDPP reaction path interpolation, and automated directory generator.
11. **1 1 Phonons:** Finite displacement frozen-phonon matrix builder (`FC.first` , `FC.last`) and Vibra dispersion runner.
12. **1 2 Surface & Work Function Studio:** Planar average charge density difference ($\Delta\rho(z)$) integrator, multi-surface work function comparison studio, and 3D heterojunction band alignment diagram generator.
13. **1 3 VASP to FDF Converter:** Isolated converter that ingests VASP `POSCAR` / `CONTCAR` files with ASE GUI `Selective dynamics` and outputs complete `%block Geometry.Constraints` blocks.

Step-by-Step Hands-On Tutorials

Tutorial 1: Bulk Silicon (Si) Diamond — Ground State, Bands & DOS

Objective: Optimize diamond-structure Silicon, compute the SCF ground state, band structure, and electronic density of states.

Complete Input File (`silicon.fdf`)

```
 SystemLabel          silicon
SystemName           Bulk Silicon Diamond
NumberOfAtoms        8
NumberOfSpecies      1

%block ChemicalSpeciesLabel
  1   14  Si      Si.psf
%endblock ChemicalSpeciesLabel

LatticeConstant      5.4307 Ang
%block LatticeVectors
  1.000000  0.000000  0.000000
  0.000000  1.000000  0.000000
  0.000000  0.000000  1.000000
%endblock LatticeVectors

AtomicCoordinatesFormat Fractional
%block AtomicCoordinatesAndAtomicSpecies
  0.000000  0.000000  0.000000  1
  0.500000  0.500000  0.000000  1
  0.000000  0.500000  0.500000  1
  0.500000  0.000000  0.500000  1
  0.250000  0.250000  0.250000  1
  0.750000  0.750000  0.250000  1
  0.750000  0.250000  0.750000  1
  0.250000  0.750000  0.750000  1
%endblock AtomicCoordinatesAndAtomicSpecies

# Basis and Grid
PAO.BasisSize        DZP
PAO.EnergyShift      100.0 meV
MeshCutoff           300.0 Ry
XC.functional        GGA
XC.authors           PBE

# SCF Parameters
MaxSCFIterations     100
DM.Tolerance         1.0d-4
DM.MixingWeight      0.25
DM.NumberPulay       8

# K-Points
%block kgrid_Monkhorst_Pack
  6   0   0   0.5
  0   6   0   0.5
  0   0   6   0.5
%endblock kgrid_Monkhorst_Pack

# Band Structure Path
BandLinesScale ReciprocalLatticeVectors
%block BandLines
  1   0.000  0.000  0.000  \Gamma
 30   0.500  0.000  0.500  X
 15   0.500  0.250  0.750  W
 20   0.500  0.500  0.500  L
 30   0.000  0.000  0.000  \Gamma
%endblock BandLines

# Output Flags
```

```
WriteMullikenPop     1
WriteForces          true
SaveRho              true
```

Execution and Analysis

```
        # Run SIESTA with MPI:
        mpirun -np 4 siesta < silicon.fdf | tee silicon.out

        # Extract band structure:
        gnubands silicon.bands > bands.dat
```

Tutorial 2: $FAPbI_3$ Perovskite (001) Slab — Cleaving, Layer Freezing & Work Function

Objective: Construct a polar organic-inorganic perovskite slab, apply 20 Å vacuum, freeze the bottom two substrate layers to mimic semi-infinite bulk, activate slab dipole corrections, and compute the work function Φ.

Workflow Steps in the GUI

1. In **2 Structure**, click **Import CIF** and select `FAPbI3_cubic.cif`.
2. Click **Add Vacuum** and specify `20.0 Å`. The cell vector c automatically expands to 36.0 Å, and `Slab.DipoleCorrection true` is activated.
3. Under **Atomic & Layer Constraints**, freeze the bottom atoms ($z < 15$ Å) while keeping the surface atoms free to reconstruct.
4. Set `kgrid_Monkhorst_Pack` to $4 \times 4 \times 1$.
5. Turn on `SaveElectrostaticPotential true` in **8 Output Flags**.
6. Export `FAPbI3_slab.fdf`.

Complete Input File (`FAPbI3_slab.fdf`)

```
 SystemLabel          FAPbI3_slab
SystemName           FAPbI3 Perovskite (001) Surface Slab
NumberOfAtoms        20
NumberOfSpecies      5

%block ChemicalSpeciesLabel
  1   82  Pb      Pb.psf
  2   53  I       I.psf
  3    6  C       C.psf
  4    7  N       N.psf
  5    1  H       H.psf
%endblock ChemicalSpeciesLabel

LatticeConstant      1.0 Ang
%block LatticeVectors
  12.640000   0.000000   0.000000
   0.000000  12.640000   0.000000
   0.000000   0.000000  36.000000
%endblock LatticeVectors

AtomicCoordinatesFormat NotScaledCartesianAng
AtomicCoordinatesOrigin COP-Z
%block AtomicCoordinatesAndAtomicSpecies
  # Bottom Substrate Layer (Fixed)
  0.0000   0.0000   8.2000  1   # Pb [F F F]
  6.3200   6.3200   8.2000  2   # I  [F F F]
  0.0000   6.3200   8.2000  2   # I  [F F F]
  6.3200   0.0000   8.2000  2   # I  [F F F]
  # Surface Reconstructed Layer (Free)
  0.0000   0.0000  14.5200  1   # Pb [T T T]
  6.3200   6.3200  14.5200  2   # I  [T T T]
  # Molecular Cations (FA: CH(NH2)2+)
  3.1600   3.1600  11.3600  3   # C  [T T T]
  3.1600   2.1000  11.3600  4   # N  [T T T]
  3.1600   4.2200  11.3600  4   # N  [T T T]
%endblock AtomicCoordinatesAndAtomicSpecies

# Constraints (Section 7.7)
%block Geometry.Constraints
  atom [1 -- 4]              # Freeze bottom Pb-I layer
  cell-vector C              # Lock vacuum length
%endblock Geometry.Constraints

# Slab Physics
Slab.DipoleCorrection  true
DFTD3                  true
PAO.BasisSize          DZP
MeshCutoff             350.0 Ry
XC.functional          GGA
XC.authors             PBE

# 2D Surface K-Points (kz = 1)
%block kgrid_Monkhorst_Pack
  4   0   0   0.5
  0   4   0   0.5
  0   0   1   0.0
%endblock kgrid_Monkhorst_Pack

# Relaxation & Potential Output
```

```
MD.TypeOfRun           CG
MD.NumCGsteps          150
MD.MaxForceTol         0.04 eV/Ang
SaveElectrostaticPotential true
```

Extracting the Work Function

Upon completion, the electrostatic potential is stored in `FAPbI3_slab.VH`. Convert to planar average using Macroave:

```
        macroave < macroave.in
```

Upload the generated `planar_potential.dat` into **1 2 Surface & Work Function Studio**. The GUI plots $\bar{V}(z)$, detects the flat vacuum level ($V_{\text{vac}} = 0.30$ eV), reads the Fermi level ($E_F = -4.35$ eV), and computes:

$$\Phi = 0.30 \text{ eV} - (-4.35 \text{ eV}) = 4.65 \text{ eV}$$

Tutorial 3: Gas Adsorption on Surface — Energy & Charge Density Difference

Objective: Calculate the binding energy E_{ads} of a water molecule (H_2O) on a TiO_2 anatase surface, and compute the planar charge transfer $\Delta\rho(z)$.

The Three Independent Calculations

To compute the adsorption energy:

$$E_{\text{ads}} = E_{\text{total}}[\text{Slab + Mol}] - E_{\text{slab}}[\text{Clean Slab}] - E_{\text{mol}}[\text{Isolated Mol}]$$

1. **System 1 (Total):** H_2O adsorbed on TiO_2 slab in cell $a \times b \times c$. Output: $E_{\text{total}} = -14256.789$ eV.
2. **System 2 (Clean Slab):** Bare TiO_2 slab in the exact same cell $a \times b \times c$. Output: $E_{\text{slab}} = -13850.123$ eV.
3. **System 3 (Free Molecule):** Isolated H_2O molecule in the exact same cell. Output: $E_{\text{mol}} = -405.892$ eV.

$$E_{\text{ads}} = -14256.789 - (-13850.123) - (-405.892) = -0.774 \text{ eV} \quad \text{(Exothermic Physisorption)}$$

Planar Average Charge Density Difference (CDD)

Request valence charge densities: `SaveRho true`. Generate 3D grid files:

$$\Delta\rho(\mathbf{r}) = \rho_{\text{total}}(\mathbf{r}) - \rho_{\text{slab}}(\mathbf{r}) - \rho_{\text{mol}}(\mathbf{r})$$

Integrate in the xy-plane to yield the planar profile:

$$\Delta\bar{\rho}(z) = \frac{1}{A}\iint \Delta\rho(x,y,z)\,dx\,dy$$

and the cumulative charge transfer across the interface:

$$\Delta Q(z) = \int_{-\infty}^{z} \Delta\bar{\rho}(z')\,dz'$$

In **1 2 Surface & Work Function Studio → Planar Average CDD**, uploading the resulting `.cube` or `.dat` file renders the dual-axis chart:

- Peak accumulation region (+$\Delta\rho$: electron buildup at the interface bond).
- Peak depletion region (−$\Delta\rho$: charge loss from the substrate surface).
- Net transferred charge Δq and interface dipole $\Delta\mu$ in Debye.

End of Documentation

Generated for SIESTA 5.x & SIESTA Input Builder Web Studio